Two-channel signatures of the E_7 spectrum in a blockaded Rydberg chain

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The thermal deformation of the tricritical Ising conformal field theory is an integrable theory with seven E_7 particles. In a blockaded Rydberg chain, lattice momentum separates its odd and even particles: a staggered detuning probe couples to σ and a uniform one to ε. Exact diagonalization of the strictly blockaded chain with a $1/r^6$ tail, up to $L = 36$, locates a direction-dependent parameter region in which the four light particles A_1–A_4 appear as two sharp peaks per channel, with gap ratios within 3% of E_7. Spectral weights distinguish these single-particle peaks from scattering levels, whose weights fall with system size. Shifting the interaction by its reported 1.7% calibration uncertainty moves each ratio by less than half a percentage point. On open chains, reflection symmetry makes the two probes exactly orthogonal at even length, whereas odd chains of 19–27 atoms leave up to 7.5% leakage. The shallow bound state A_4 is the least converged: its binding energy, $2m_1 - m_4$, still decreases with L and remains about 50% above E_7 after extrapolation. The heavy particles A_5–A_7 are not resolved.

Introduction.—Relevant perturbations of a two-dimensional conformal field theory (CFT) can produce integrable massive theories with spectra fixed by exceptional Lie algebras. The magnetic perturbation of critical Ising theory realizes E_8 [1], with experimental signatures in $CoNb_2O_6$ [2] and Rydberg arrays [3]. The leading thermal perturbation of the tricritical Ising (TCI) CFT, with central charge $c = 7/10$, instead realizes E_7 [4–6]. Writing $r_a = m_a/m_1$, the seven ratios are

$$(r_1, \ldots, r_7) = (1,\ 1.2856,\ 1.8794,\ 1.9696,\ 2.5321,\ 2.8794,\ 3.7017). \quad (1)$$

with $\mathbb{Z}_2$ parities $(-,+,-,+,+,-,+)$ in the disordered phase [7]. An experimental identification of this spectrum remains an open problem [8].

Rydberg chains host Ising criticality, and an effective attraction produces a tricritical point [9, 10]. Sun *et al.* [11] reached it by preparing the upper edge of the blockaded sector, equivalent in the projected model to an attractive ground-state problem, and measured finite-size CFT spectra. This motivates spectroscopy of the neighboring gapped regime.

The field-theory inputs are established: the Rydberg density–CFT operator correspondence [10], the E_7 form factors and selection rules [6, 12], and their threshold structure. Király and Lencsés [13] have also obtained lattice-quench spectra compatible with the even particles A_2, A_4, A_5 in the Blume–Capel chain, including form-factor amplitude comparisons. That study uses A_2 to calibrate the mass scale; it does not independently resolve A_1. Here we show that a Rydberg chain gives access to both parity sectors, so that odd and even light particles can be compared on one platform. We map the parameter region where four light peaks match E_7, show that even open chains remove cross-channel leakage by symmetry, and quantify the finite-size constraint set by A_4.

Model and field content.—We study the projected Hamiltonian

$$H = \sum_j \left[\frac{\Omega}{2} P_{j-1} X_j P_{j+1} - \Delta n_j\right] + \sum_{i<j,\, d_{ij}\geq 2} V_2 (2/d_{ij})^6 n_i n_j. \quad (2)$$

with the constraint $n_j n_{j+1} = 0$, $P_j = 1 - n_j$, and $V_2 < 0$. The distance d_{ij} is $|i-j|$ for open chains and the shortest distance on a periodic chain; each pair is counted once. We set $\hbar = \Omega = a = 1$, where a is the lattice spacing. All allowed interaction pairs are retained within each finite chain.

Equation (2) is the strict-blockade approximation to the sign-reversed experimental problem, not the full experimental Hamiltonian. It omits the finite-blockade correction H_2 of Ref. [11], which contains diagonal shifts and correlated hopping of order Ω^2/V_1. We compute ground-state response on open chains without boundary detuning.

In the long-wavelength theory, one-site translation exchanges the two density-wave patterns and implements the internal $\mathbb{Z}_2$ together with spatial translation [10]. The density has the expansion

$$n_j \simeq \bar{n} + c_\varepsilon \varepsilon + c_{\varepsilon'} \varepsilon' + (-1)^j (c_\sigma \sigma + c_{\sigma'} \sigma') + \cdots . \quad (3)$$

Thus zero-momentum even particles occur near lattice momentum 0, and odd particles near π. This identification uses bulk translation symmetry; open-chain reflection parity is a different quantum number.

Both Δ and V_2 generate the even relevant perturbations $g_\varepsilon \varepsilon + g_{\varepsilon'} \varepsilon'$, with RG eigenvalues 9/5 and 4/5. For a straight path of distance t from TCI with a nonzero linear component in g_ε,

$$\eta = \frac{g_{\varepsilon'}}{|g_\varepsilon|^{4/9}} \propto t^{5/9}, \qquad m_1 \propto t^{5/9}. \quad (4)$$

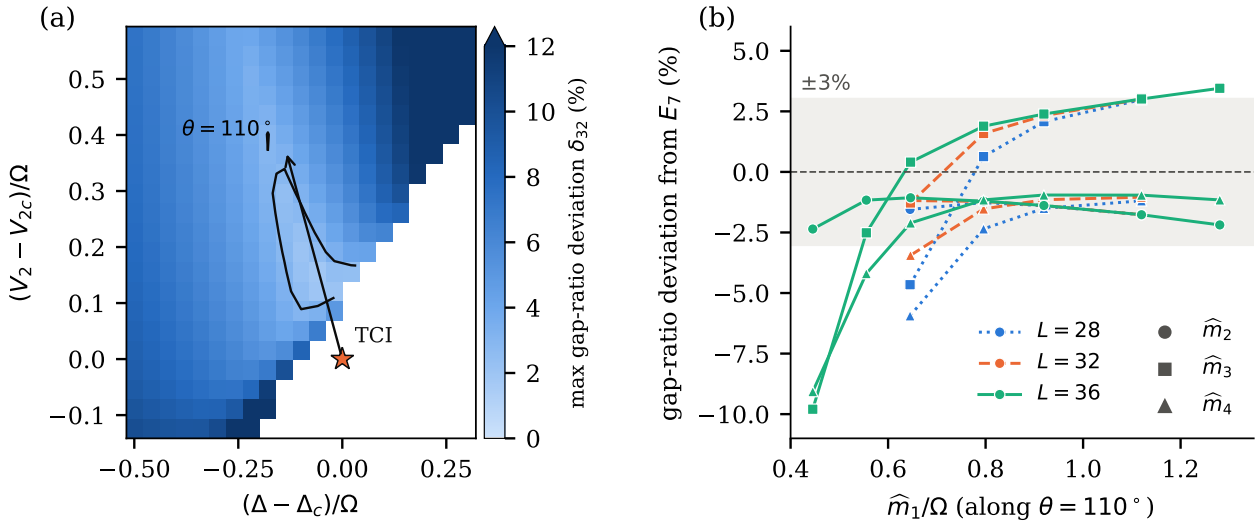


FIG. 1. Finite-chain spectral matching region. (a) $\delta_L = 100\,\max_{a=2,3,4}|(\widehat{m}_a/\widehat{m}_1)/r_a - 1|$ at $L = 32$; the contour marks 3%. Grid spacings are 0.04Ω in Δ and 0.035Ω in V_2. Blank cells fail the numerical filters $L\widehat{m}_1/\Omega \geq 18$ and $\widehat{m}_2 \geq \widehat{m}_1/2$; these filters do not establish phases or convergence. The star estimates TCI; the arrow marks $\theta = 110°$. (b) Individual gap-ratio deviations along that direction. Symbols identify the candidate levels and line styles identify sizes. The band is ±3%; lines guide the eye. Neither panel is an experimental error budget.

Such paths approach E_7 as $t \to 0$ only if L grows with the correlation length. We have not calibrated the map to $(g_\varepsilon, g_{\varepsilon'})$, so a region of small ratio errors need not be the pure thermal trajectory.

Probes and numerical method.—For a periodic ground state, the connected density response is

$$S_k(\omega) = \frac{1}{L}\sum_{n\neq 0}\left|\left\langle n\left|\sum_j e^{ikj}n_j\right|0\right\rangle\right|^2 \delta[\omega-(\mathcal{E}_n-\mathcal{E}_0)]. \quad (5)$$

Uniform and staggered probes select $k = 0$ and π exactly. In the E_7 limit, Eq. (3) couples them principally to ε and σ, respectively. The proposed $k = \pi$ coupling to σ already appears in Ref. [11]. For a pure continuum operator $\mathcal{O}$, one-particle weights obey $w_a \propto |F_a^{\mathcal{O}}|^2/m_a$. Known form factors give $w_3/w_1 = 0.067$, $w_6/w_1 = 0.0030$ for σ, and $w_4/w_2 = 0.143$, $w_5/w_2 = 0.038$, $w_7/w_2 = 0.0012$ for $\Theta \propto \varepsilon$ [6, 12]. Absolute interchannel amplitudes are not universal.

We use sparse exact diagonalization in the constrained space for periodic $L \leq 36$ and open $L = 19$–28, with reorthogonalized Lanczos spectra (up to 400 iterations); light levels agree with direct eigensolver results to 10^{-6}, and small systems match full-space diagonalization to machine precision.

We denote the first two gaps in the $k = \pi$ sector by $\widehat{m}_1(L), \widehat{m}_3(L)$ and the first two excited gaps at $k = 0$ by $\widehat{m}_2(L), \widehat{m}_4(L)$. These are candidate levels, labeled by energy within each sector; ω_a^{o} denotes the corresponding dominant open-chain resonance.

Simultaneous crossings of L times the lowest odd and lowest excited even gaps locate a finite-size TCI estimate. The $L = 28/32$ crossing gives $(\Delta_c, V_{2c}) = (-1.5011, -1.6260)$, within 10^{-3} of the 24/28 estimate and close to the experimental calibration [11]. Fixing $v \simeq 1.90$ with $x_\sigma = 3/40$, the other rescaled gaps, 0.875 ($k = \pi$) and 0.198, 1.18 ($k = 0$), match $x_{\sigma'} = 7/8$, $x_\varepsilon = 1/5$, $x_{\varepsilon'} = 6/5$.

Matching region and finite size.—We scan $(\Delta, V_2) = (\Delta_c, V_{2c}) + t(\cos\theta, \sin\theta)$. Figure 1 identifies a finite region of small errors among the three gap ratios. Along $\theta = 110°$ at $L = 36$, $t = 0.1$–0.2 give maximal errors below 3% (3.01% at $t = 0.3$); 90° and 130° also contain matching points, whereas $\theta = 180°$ never reaches 5% and 210° gives 10–25%. This direction dependence concerns the finite-chain window, not the asymptotic limit of Eq. (4).

At $\theta = 110°, t = 0.2$, corresponding to $(\Delta, V_2) = (-1.5695, -1.4381)$, we find $\widehat{m}_1(36) = 0.9196\Omega$ and

$$\frac{(\widehat{m}_2, \widehat{m}_3, \widehat{m}_4)}{\widehat{m}_1} = (1.2678,\ 1.9243,\ 1.9508). \quad (6)$$

The deviations are $(-1.39, +2.39, -0.95)\%$. Closer to TCI, at $t = 0.1$, $\widehat{m}_3/\widehat{m}_1$ still moves from 1.792 to 1.887 between $L = 28$ and 36; at $t = 0.3$ all ratios move by less than 0.24%.

The shallow bound state A_4 needs care. In E_7 it is an A_1A_1 bound state with

$$\mu_{11}^4 = \sqrt{m_1^2 - m_4^2/4} = m_1 \sin(\pi/18) \simeq 0.174 m_1. \quad (7)$$

Its finite-volume energy can receive a leading bound-state correction proportional to $e^{-\mu_{11}^4 L/v}$, not simply

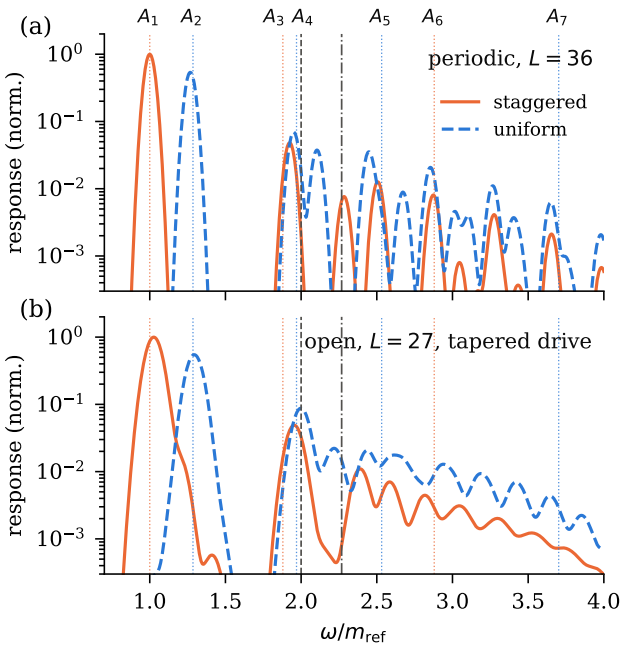


FIG. 2. Response at Eq. (6): (a) periodic $L = 36$, without size averaging; (b) open $L = 27$ with a $\sin^2$ drive envelope. Both use $m_{\rm ref} = \widehat{m}_1(36) = 0.9196\Omega$ and are normalized to their own staggered peak height. Solid orange: staggered; dashed blue: uniform. Only (a) has exact momentum selection. Gaussian standard deviations $0.03m_{\rm ref}$ and $0.05m_{\rm ref}$ illustrate smoothing, not instrumental linewidths. Dotted lines mark ideal E_7 ratios. Gray dashed/dash-dotted lines mark $2m_{\rm ref}$ and $m_{\rm ref} + \widehat{m}_2(36)$: periodic-gap sums, not measured open-chain continuum edges.

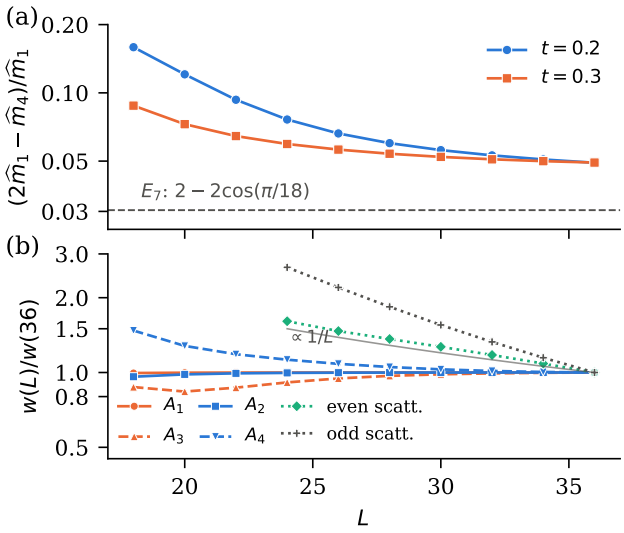


FIG. 3. Size dependence along $\theta = 110^\circ$ from periodic spectra. (a) Binding $b_4 = (2\widehat{m}_1 - \widehat{m}_4)/\widehat{m}_1$ at $t = 0.2$ (working point) and $t = 0.3$; the dashed line is E_7. (b) Weights at $t = 0.2$ relative to $L = 36$. Orange: odd; blue: even. Light particles converge; the lowest scattering levels (dotted, $L \geq 24$) decrease at least as fast as the gray $1/L$ reference.

$e^{-m_1 L/v}$ [12, 14]. With the working gap and $v \simeq 1.90$, the E_7 value of μ_{11}^4 gives $v/\mu_{11}^4 \simeq 12$ sites; the binding measured on the chain gives $\simeq 9$ sites. Both exceed $v/m_1 \simeq 2$ sites considerably. We therefore track the binding $b_4 \equiv (2\widehat{m}_1 - \widehat{m}_4)/\widehat{m}_1$ directly [Fig. 3(a)]. At the working point it falls from 0.159 at $L = 18$ to 0.0492 at $L = 36$, still decreasing, although $\widehat{m}_1$ has converged to 10^{-5} for $L \geq 28$. A fit $b_4(L) = b_4^\infty + ce^{-L/\ell}$ for $L \geq 26$ gives $b_4^\infty \simeq 0.045$, with $\ell \simeq 6$ sites and a fit-window dependence of 0.001; the E_7 value is 0.0304. The remaining $\sim 50\%$ excess is then a nonuniversal correction to the binding, although a slower finite-size component cannot be excluded. Since the 3% ratio tolerance would allow $\widehat{m}_4 > 2\widehat{m}_1$, we state b_4 separately: it is the most sensitive light-sector test of E_7.

Spectral weights identify single-particle levels. For the normalization of Eq. (5), a stable particle has an L-independent weight, whereas a scattering state at fixed total momentum carries weight $\propto 1/L$. Figure 3(b) shows both behaviors. From $L = 24$ to 36, the weights of A_1 and A_2 change by less than 0.4%, and that of A_3 by 9% with decreasing increments. The lowest scattering levels fall by 2.6 (odd) and 1.6 (even), comparable to or faster than $1/L$. The A_4 weight decreases by 11% and flattens, consistent with a bound state that is not yet converged. At $L = 36$, $w_4/w_2 = 0.129$ and $w_3/w_1 = 0.0483$, 10% and 28% below the pure-field values. Mixing with σ' contributes an effective relative amplitude proportional to $(c_{\sigma'}/c_\sigma)(m_1 a/v)^{4/5}$ at fixed CFT normalization, since $x_{\sigma'} - x_\sigma = 4/5$. At $L = 36$ along the cut, w_3/w_1 rises from 0.034 at $t = 0.4$ to 0.052 at $t = 0.1$, toward the pure-σ value, while w_4/w_2 crosses 0.143 near $t = 0.15$ and overshoots closer to TCI, where finite size dominates. Interference and finite-size corrections are therefore not separated here.

Spectra and boundaries.—The periodic response in Fig. 2(a) has two prominent subthreshold peaks in each sector, consistent with the four candidate levels in Eq. (6). Momentum selection is exact. The next even level, at $2.105m_{\rm ref}$, lies well above $2m_{\rm ref}$, so a finite periodic spectrum overstates the gap between A_4 and the continuum.

For open chains we use $O_{f,q} = \sum_j f_j e^{iqj} n_j$, where $q = 0, \pi$ labels the drive pattern rather than an exact momentum sector. Spectral weights are normalized by $\sum_j f_j^2$. We compare $f_j = 1$ with $f_j = \sin^2[\pi(j+1)/(L+1)]$, $j = 0, \ldots, L-1$. At $L = 27$, the latter gives dominant candidate resonance ratios

$$(\omega_2^{\rm o}, \omega_3^{\rm o}, \omega_4^{\rm o})/\omega_1^{\rm o} = (1.2612,\ 1.9030,\ 1.9488), \qquad (8)$$

with $\omega_1^{\rm o} = 0.9438\Omega$, within 2% of E_7. The open-chain energies are shifted upward by the lowest standing-wave momentum. With $E_a(n) = \{\widehat{m}_a^2 + [nv\pi/(L+1)]^2\}^{1/2}$, $v = 1.90$, and the periodic $L = 36$ masses, the $n = 1$ prediction reproduces $\omega_1^{\rm o}$ to 0.02% and $\omega_{2,3}^{\rm o}$ to better

than 0.8% at $L = 27$. The largest deviation occurs for A_4: 1.8%, 2.9%, and 5.2% at $L = 27$, 23, and 19. This size dependence is consistent with A_4 having the largest spatial extent, as estimated above.

Reflection $R : n_j \to n_{L-1-j}$ fixes the leakage structure. For symmetric envelopes, $RO_{f,0}R^{-1} = O_{f,0}$ and $RO_{f,\pi}R^{-1} = (-1)^{L-1}O_{f,\pi}$. At even L the two probes therefore create states of opposite reflection parity and cannot excite the same eigenstate. At odd L both probes are reflection even. Table I confirms this: the same-state leakage, defined as the weight at a candidate eigenenergy in the unintended probe relative to the intended probe, vanishes to machine precision for $L = 26, 28$. At odd L it remains up to 7.5% for A_3 at $L = 27$ even with tapering, which reduces A_1 from 0.15% to 0.027% but raises A_4 from 1.4% to 1.9%.

Reflection parity is not the internal $\mathbb{Z}_2$, so even chains remove leakage, not the need for resolution: at $L = 28$ the tapered A_3 and A_4 peaks remain $0.039\,\omega_1^{\circ}$ apart in the two probes, and the A_2 response splits into lines at 1.250 and $1.266\,\omega_1^{\circ}$ with weights 27 : 73. The strongest staggered satellite, at $1.160\,\omega_1^{\circ}$ for $L = 27$, is consistent with the $n = 3$ standing wave of A_1: $(L+1)(E^2-\widehat{m}_1^2)^{1/2}$ is 16.4–16.6 for $L = 19$–27, 2.7–2.8 times the $n = 1$ value, and its flat-drive weight (9–13% of A_1) drops to 1.7% with the taper.

TABLE I. Same-state leakage in percent for tapered open-chain probes at the working point. The intended probe is staggered for A_1, A_3 and uniform for A_2, A_4. Even-L values vanish by reflection symmetry (below 10^{-25}).

Candidate	$L = 19$	23	27	26	28
A_1	0.228	0.074	0.027	0	0
A_2	4.53	0.009	0.019	0	0
A_3	39.4	17.8	7.46	0	0
A_4	8.98	4.42	1.94	0	0

Thresholds and unresolved heavy particles.—In the integrable, translation-invariant theory, the lowest even and odd two-particle thresholds are $2m_1$ (A_1A_1) and m_1+m_2 (A_1A_2). The exact E_7 scattering amplitudes and form factors imply responses proportional to $(\omega-2m_1)^{1/2}$ and $(\omega-m_1-m_2)^{-1/2}$ in the corresponding channels [12]. Here $S_{11}(0) = -1$ forces the identical-particle form factor to vanish, whereas the relevant A_1A_2 form factor is nonzero at threshold. These are integrable-limit predictions for the matrix-element-weighted response.

At the working point, the lowest odd scattering level moves from 2.340 to $2.284\widehat{m}_1$ between $L = 18$ and 36, toward $1 + \widehat{m}_2/\widehat{m}_1 = 2.268$, and the even one from 2.296 ($L = 24$) to $2.105\widehat{m}_1$, toward 2. This is consistent with the two onsets but does not determine the exponents.

We do not resolve A_5, A_6, A_7. In the E_7 theory their masses lie above allowed thresholds in their own sectors: A_5, A_7 above $2m_1$, and A_6 above $m_1 + m_2$. Nonintegrable perturbations can open decay channels such as $A_5 \to A_1A_1$ and $A_6 \to A_1A_2$ [7]. Across $L = 18$–36 at $\theta = 110^{\circ}$, $t = 0.2, 0.3$, the higher spectra do not provide a stable heavy-particle assignment by energy and weight. Finite chains do not measure decay widths, so this neither establishes nor excludes heavy resonances in larger systems.

Experimental requirements.—The protocol prepares a state near Eq. (6) with the demonstrated TCI preparation [11], applies weak tapered uniform and staggered modulations, and compares the two responses. An even number of atoms is preferable, since it removes same-state leakage. The sign-changing staggered drive needs the two-color (515 and 813 nm) tweezer scheme proposed in Ref. [11], beyond the one-color light shifts used so far.

Spectral resolution is especially consequential for A_4. The finite-chain estimate $2\widehat{m}_1 - \widehat{m}_4 \simeq 0.049m_{\rm ref}$ is about $2\pi \times 0.25$ MHz for $\Omega = 2\pi \times 5.5$ MHz; in ideal E_7 the relative binding is smaller, $0.0304m_1$. The illustrative open-chain broadening in Fig. 2 has FWHM $= 2\sqrt{2\ln 2}\,\sigma = 0.118m_{\rm ref}$, about $2\pi \times 0.60$ MHz. Separating A_4 from the continuum therefore needs a linewidth below this scale; the E_7 threshold suppression of the A_1A_1 response would help in the scaling regime. At odd L, mixing means that A_3 and A_4 should be assigned by comparing both probes rather than by channel label alone.

To test calibration, we recompute the working point with V_2 scaled by 1 ± 0.017, the reported uncertainty [11]. At $L = 36$, $\widehat{m}_1$ shifts by 4.3–4.5%, but the ratios change by at most 0.12, 0.43, and 0.20 percentage points for $a = 2, 3, 4$ ($L = 32$ is similar). The binding b_4 changes from 0.0492 to 0.0461 and 0.0530, by 6–8%. Because ratios are taken at fixed parameters, this calibration error mainly rescales m_1 and is not the limiting systematic.

Finite-blockade corrections, preparation, decoherence, and readout are not modeled; they must be included before the four-peak signature is used as an experimental identification.

Conclusion.—Because one-site translation implements the tricritical $\mathbb{Z}_2$, a Rydberg chain can resolve both parity sectors of the E_7 spectrum with uniform and staggered probes. In the strictly blockaded model, four light particles appear as two peaks per channel at E_7 ratios within 3% in a direction-dependent region. Their weights separate them from scattering levels, and the ratios are insensitive to the reported calibration uncertainty. Even-length open chains make the two probes orthogonal by reflection symmetry. The shallow bound state A_4 is the least converged and sets the resolution requirement; the heavy particles remain open.

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